Layover Compensation Method for Regional Spaceborne SAR Imagery Without GCPs

Huabin Wang, Qian Cheng, Taoyang Wang, Guo Zhang, Yunming Wang, Xin Li, and Boyang Jiang

***Abstract*— Synthetic aperture radar (SAR) images reveal severe geometric distortions especially in the mountain area, such as layover, which is caused by imaging characteristics of SAR itself and terrain undulations. The layover phenomenon greatly limits the application of SAR images. This article proposes a layover compensation method for regional spaceborne SAR imagery without ground control points (GCPs), which is mainly improved from two aspects. First, a method based on rational function model (RFM) to determine the layover range is proposed. Second, based on geometric calibration and block adjustment, the processing flow is optimized to generate digital orthophoto map (DOM) which greatly eliminated the influence of layover. The proposed method was applied to Chinese Gaofen-3 (GF-3) SAR regional images, including the ascending and descending track stacks. The result showed that 84.5% of the layover pixels on the regional DOM were compensated, which verified the effectiveness and feasibility of the method.**

***Index Terms*— Block adjustment, GF-3, layover compensation, synthetic aperture radar (SAR).**

I. Introduction

SPACEBORNE synthetic aperture radar (SAR) sensors are being continuously developed globally, and the geometric positioning accuracy and resolution of SAR sensors have increased significantly in recent years. At present, the positioning accuracy of the Sentinel-1A strip-map mode reaches the sub-pixel level [1], and the COSMO-SkyMed satellite in Spotlight-2 mode achieves a resolution of up to 1 m [2]. Unlike optical remote sensing, spaceborne SAR, a type of active microwave remote sensing technology, has all-weather imaging capabilities that have been widely used in surveying and mapping [3], geology [4], oceanography [5], [6], and other fields. However, due to the side-looking and range imaging characteristics, there is geometric distortion in the SAR imagery, such as layover especially in the mountainous regions. Because of the prevalence of layover in mountainous areas, scattering signals from these distortion regions hardly contain information about terrain cover types [7]. Therefore, the layover phenomenon greatly limits the application of SAR images.

The principle of layover compensation is that when the pixel is a layover pixel on master image, if the corresponding position on the slave image is a normal pixel, the layover pixel on the master image will be replaced by the normal pixel on the slave image. When the layover pixels on the image are masked out, the remaining pixels are regarded as normal pixels. At present, relevant scholars have used SAR for production of regional digital orthophoto map (DOM). Wang *et al.* [8] used Yaogan-5 (YG-5) images to generate regional orthophoto maps in Xi'an, although the absolute positioning accuracy was better than 1 pixel. However, in the process of DOM production, no ascending or descending image was used to compensate for layover. Zhang *et al.* [9] used GF-3 satellite Fine Strip II (FSII) mode SAR images to generate the Hubei DOM; however, layover pixels in single image still exist.

To solve the problem of layover on the SAR DOM, this article proposes a layover regional compensation method for regional spaceborne SAR imagery without GCPs. Using ascending and descending regional images of GF-3 in quad-polarization strip I (QPSI)-mode, Master Image Data (MID) and Slave Image Data (SID) are selected. The refined RFM is obtained by geometric calibration and block adjustment. Then, orthorectification is carried out and Layover Mask Image (LMI) is generated. Next, layover compensation is carried out. Finally, we completed compensated regional SAR DOM by mosaic.

II. Related Work

The research about layover compensation is mostly based on the range–Doppler (RD) model. Wan *et al.* [10] used a two-step process to perform geometric corrections of

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Radarsat-2 ascending and descending images. The results of an experiment in layover compensation showed that an absolute positioning accuracy of 5 m and a rate of compensation better than 90% were achieved. Then, a simulated image generated using a digital elevation model (DEM) was found to improve the positioning accuracy. The results of a compensation experiment conducted using Radarsat-2 images showed that more than half of the area was compensated [11]. Combining an RD model with sparse GCPs to perform orthorectification, Zhang *et al.* [7] adopted Radarsat-2 dual-aspect images to perform layover compensation. The experiment yielded the desired results.

Layover compensation has two important parts, including the high-precision positioning of the original SAR image and the generation of LMI. In geometric processing of spaceborne SAR imagery, the RD model is rigorous and widely used [12], [13]. Because an iterative solution process is associated with the RD model, the problem of slow solution speeds becomes increasingly prominent [14]. RFM is a general geometric processing model that has been widely used in optical satellite image. It has good interpolation characteristics continuity; besides, it is independent of the sensor and platform. Zhang *et al.* [15] proposed that RFM could be used to replace the RD model, and it has increasingly been applied to geometric correction of spaceborne SAR data.

Rational polynomial coefficients (RPCs) are model parameters of RFM, which are affected by spaceborne SAR system error and need to be optimized. Eftekhari *et al.* [16] used GCPs measured by GPS to conduct four-corner control for TerraSAR-X data and achieved an accuracy better than 1 pixel. Wang *et al.* [13] refined the RPCs and achieved positioning accuracy better than 1.5 pixels for five GF-3 images. Zhang *et al.* [17] used RFM to generate simulated images for mountainous areas and carried out orthorectification, and the positioning accuracy was better than 7 m.

In addition to using GCPs to modify model parameters for each SAR image, geometric calibration can also be used to improve the geometric accuracy. Generally, once after calibration, it can ensure SAR image to achieve high positioning accuracy without GCPs in quite a long time. Considering atmospheric propagation delay, Zhao *et al.* [18] proposed a multimode hybrid geometric calibration method. Zhao *et al.* [19] then conducted experiments using GF-3 satellite imagery, and the result showed that the systemic errors of the GF-3 SAR satellite could be effectively eliminated and a geometric accuracy of 3 m could be achieved by calibration. Deng *et al.* [20] proposed a geometric cross-calibration method for GF-3 imagery that does not rely on corner reflectors and high-precision DEM. Compared with the method of using GCPs, geometric calibration method is more promising in situations in which GCPs are difficult to acquire.

So far, there are not many researches on the generation of layover mask. In the literature [21], the layover area can be detected by computing for each DEM sub-sample the magnitude of SAR look angle and slant range. Zhang *et al.* [7] proposed a layover area determination method based on the relationship between the look angle and the slope angle in the range direction. However, both the methods require calculation of the corresponding position of the satellite to solve for variables such as the slant range or the look angle. In the Gamma software, a threshold is used to determine the layover [22]. Usually, several times of the image average value are selected as threshold. However, the mask in image space may severely discretize after orthorectification.

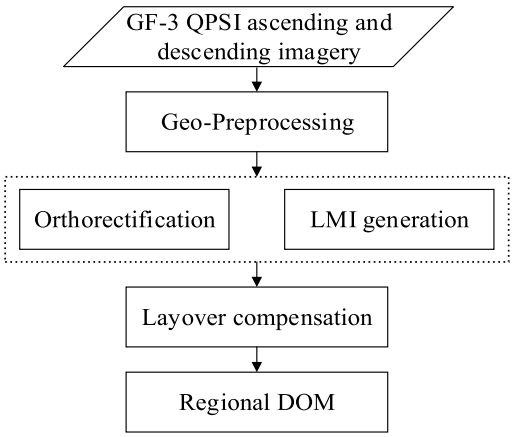


Fig. 1. Layover compensation flowchart of regional spaceborne SAR imagery.

III. Methods

In this article, first, SID, MID, and the refined RPCs are obtained by geo-preprocessing. Second, orthorectification is carried out and LMI is generated automatically. Third, layover compensation is carried out for MID. Fourth, regional SAR DOM after compensation is obtained by mosaic. The flowchart is shown in Fig. 1.

A. Geo-Preprocessing

1) Master–Slave Image Selection: A master–slave image is determined by the size of the layover area or the angle of incidence; it is determined as master image if the layover area is smaller or the incidence angle is larger. However, for regional images, it may cause overlapping images in one place and gaps in others. For example, if image set {A1, A2, C1, D1} is selected as MID, there are two "holes" in Fig. 2(b). If image set {A1, A2, C1, B1} is selected as MID, there is one "hole" in Fig. 2(c). If image set {A1, A2, B1, B2} is selected as MID, there is no "hole" in Fig. 2(d). Therefore, in this article, the full coverage images of the same viewing are selected as MID, and the other images are regarded as SID.

When MID is selected, one image in SID may overlap with multiple images in MID. For example, C1 overlaps with {A1, A2, B1, B2} in Fig. 2(a). The overlapping rate between C1 and A1 is small in Fig. 2(a), and its influence is negligible. Therefore, we do not conduct compensation between C1 and A1. Fig. 3 shows the overlapping rate calculation flowchart. *Img_A* and *Img_B* are the images in MID and SID, respectively, and *RPC_A* and *RPC_B* are the corresponding RPCs. First, we project *Img_B* four-corner points P_i (x_i, y_i) onto the Z elevation surface using *RPC_B*

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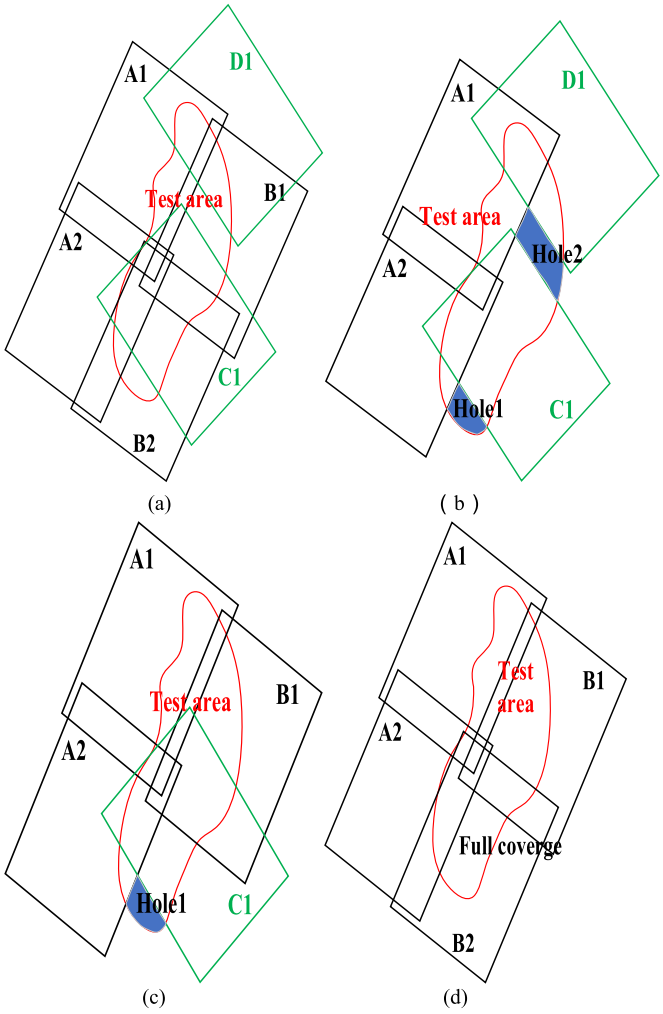


Fig. 2. Images geometric relationship. Red boundary represents the study area, black and green boundary represent the descending images and ascending images, respectively. Descending image set {A1, A2, B1, B2} cover study area fully, and ascending image set {C1, D1} cover study area partly. (a) Case 1: All MID and SID geometric relationship. (b) Case 2: Two "holes". (c) Case 3: One "hole". (d) Case 4: No "hole".

to obtain (B_i, L_i, Z), where $i = 1, 2, 3, 4$. Then, (B_i, L_i, Z) is calculated by *RPC_A* to obtain P_i' $(x_i$', y_i'). Finally, the *Img_A* four-corner points pt_i (X_i, Y_i) and P_i'$(x_i$', y_i') are used to calculate the overlapping rate. Layover compensation is performed with overlapping rate greater than threshold T_{overlap} ($T_{\text{overlap}} = 20\%$ in this article). The overlapping rate is defined as

$$\text{Rate}_{\text{overlap}} = \frac{S_{\text{overlap}}}{S_{\text{master}}} \times 100\% \tag{1}$$

where $\text{Rate}_{\text{overlap}}$ is the overlapping rate, S_{overlap} is the number of pixels in the overlapping area, and S_{master} is the number of pixels in the master image.

2) Geometric Calibration: Signal propagation delay, terrain undulations, processor error, and sensor and platform instability are the main factors that affect the accuracy of spaceborne SAR geometric positioning [23]. Depending on the characteristics of errors, they can be classified as fixed system errors, time-varying system errors, or random errors [18], [19].

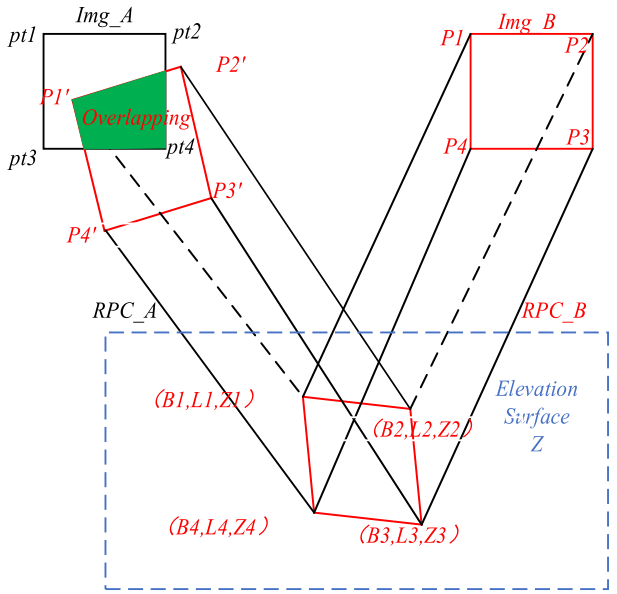


Fig. 3. Overlapping rate calculation method based on the RPC model.

The main error sources are the range and azimuth time errors. The geometric calibration model is constructed as follows:

$$\begin{cases} t_f = \left(t_{f0} + t_{\text{delay}} + \Delta t_f\right) + \dfrac{x}{f_s} \\ t_s = (t_{s0} + \Delta t_s) + \dfrac{y}{f_p}, \end{cases} \quad x \in [0, \text{width} - 1],\ y \in \left[0, \text{height} - 1\right] \tag{2}$$

where t_f and t_s are the fast time in terms of range and the slow time in terms of azimuth, respectively; t_{f0} and t_{s0} are the measured starting time values in the range and azimuth directions, respectively; t_{delay} is the atmospheric propagation time; Δt_f and Δt_s are the system delay errors; f_p is the sampling frequency; f_s is the pulse repetition; x and y are the image coordinates; and width and height are the image size in the range and azimuth directions, respectively.

The starting time of the satellite record is the time at which the radar signal is received. However, SAR satellites transmit and receive radar signals continuously, so the equivalent SAR imaging time is the intermediate time between the transmitting and receiving times [24]. Therefore, the compensation should be performed using the following equation:

$$t_{s0} = t'_{s0} - \frac{N/f_p + t_{\text{sample__delay}}}{2} \tag{3}$$

where t'_{s0} is the echo receiving time record on the satellite, N is the number of times the radar signal is transmitted from the transmitter to the receiver, and $t_{\text{sample__delay}}$ is the sample time delay in the satellite record.

Combining the spaceborne SAR geometric calibration model with the RD model, the close relationship between the object coordinates and the image coordinates of the ground

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target is constructed as follows:

$$\begin{cases} R^2 = (X_t - X_s)^2 + (Y_t - Y_s)^2 + (Z_t - Z_s)^2 \\ f_{dc} = -\dfrac{2}{\lambda R}(R_s - R_t) \times (V_s - V_t) \\ R = \left(t_{f0} + t_{\text{delay}} + \dfrac{x-1}{f_s}\right) \times c \end{cases} \tag{4}$$

where c is the speed of light; f_{dc} is the Doppler centroid frequency; λ is the radar wavelength; R is the distance from the satellite to the target point; $R_s = (X_s \quad Y_s \quad Z_s)^T$ is the satellite position vector, and V_s is the velocity vector of satellite. $R_t = (X_t \quad Y_t \quad Z_t)^T$ and V_t are the position vector and the velocity vector of the target point, respectively.

Finally, geometric calibration parameters can be calculated using GCPs by the least square method.

3) Block Adjustment Based on RFM: The RFM relates the image coordinates (r, c) to the object coordinates (X, Y, Z) in the form of the ratio of two cubic polynomials, as shown in (5). The coordinates are usually normalized to a $[-1, 1]$ range using offset and scale factors to improve the numerical precision [25], [26].

$$\begin{cases} r = \dfrac{\text{Num}_L(X, Y, Z)}{\text{Den}_L(X, Y, Z)} \\ c = \dfrac{\text{Num}_S(X, Y, Z)}{\text{Den}_S(X, Y, Z).} \end{cases} \tag{5}$$

In these equations, (X, Y, Z) are the normalized latitude, longitude, and height, respectively; and $\text{Num}_L(P, L, H)$, $\text{Den}_L(P, L, H)$, $\text{Num}_s(P, L, H)$, and $\text{Den}_s(P, L, H)$ are the terms of the third-order polynomial of (X, Y, Z). For example, $\text{Num}_L(P, L, H)$ is

$$\begin{aligned} \text{Num}_L(X, Y, Z) = {} & a_{i0} + a_{i1}Z + a_{i2}Y + a_{i3}X + a_{i4}ZY \\ & + a_{i5}ZX + a_{i6}YX + a_{i7}Z^2 + a_{i8}Y^2 \\ & + a_{i9}X^2 + a_{i10}ZYX + a_{i11}Z^2Y + a_{i12}Z^2X \\ & + a_{i13}Y^2Z + a_{i14}Y^2X + a_{i15}ZX^2 \\ & + a_{i16}YX^2 + a_{i17}Z^3 + a_{i18}Y^3 + a_{i19}X^3 \end{aligned} \tag{6}$$

where $a_{ij} (i = 1, 2, 3, 4; j = 0, 1, \ldots, 19)$ are RPCs. There are 80 parameters.

Based on the analysis of the error characteristics [27], SAR image orientation error is generally believed to be caused by the offset and scaling in the range and azimuth directions. Therefore, an affine transformation model in image space is used for error reduction [28]

$$\begin{cases} \Delta r = e_0 + e_1 r + e_2 c \\ \Delta c = f_0 + f_1 r + f_2 c \end{cases} \tag{7}$$

where $(\Delta r, \Delta c)$ is the correction for the image coordinate. (e_0, e_1, e_2) and (f_0, f_1, f_2) are the parameters of affine transformation.

Using (5) and (7), the parameters of transformation and object coordinates (X, Y, Z) are set as unknowns. The error equation can be constructed as follows:

$$\boldsymbol{V} = \boldsymbol{At} + \boldsymbol{Bx} - \mathbf{l} \tag{8}$$

where $\boldsymbol{V}$ is the residual vector of the image coordinate observation, $\boldsymbol{t} = [\Delta e_0 \ \Delta e_1 \ \Delta e_2 \ \Delta f_0 \ \Delta f_1 \Delta f_2]^T$ are the corrections for the affine transformation parameters, and $\boldsymbol{x} = [\Delta X \ \Delta Y \ \Delta Z]^T$ is the ground point incremental vector

$$\boldsymbol{A} = \begin{bmatrix} \dfrac{\partial r}{\partial e_0} & \dfrac{\partial r}{\partial e_1} & \dfrac{\partial r}{\partial e_2} & 0 & 0 & 0 \\ 0 & 0 & 0 & \dfrac{\partial c}{\partial f_0} & \dfrac{\partial c}{\partial f_1} & \dfrac{\partial c}{\partial f_2} \end{bmatrix}$$

$$\boldsymbol{B} = \begin{bmatrix} \dfrac{\partial r}{\partial X} & \dfrac{\partial r}{\partial Y} & \dfrac{\partial r}{\partial Z} \\ \dfrac{\partial c}{\partial X} & \dfrac{\partial c}{\partial Y} & \dfrac{\partial c}{\partial Z} \end{bmatrix}.$$

The solution is solved by least squares iteration until convergence. Then, RFM is refined using affine transformation parameters.

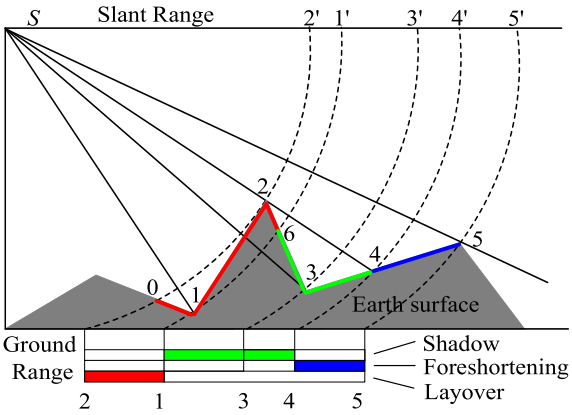


Fig. 4. Illustration of SAR imagery geometry distortion caused by terrain [31].

B. Automatic Determination of Layover Area

As shown in Fig. 4, foreshortening occurs when the imaging plane distance (slant range) between 4′ and 5′ is smaller than the distance on the actual slope surface from 4 to 5. When this happens, the energy on the slope of a certain area is imaged and compressed to a relatively small plane, so the slope facing the radar looks much brighter than its surrounding area. The degree of foreshortening is related to the slope and the incidence angle. For different SAR satellites, such as GF-3, TerraSAR-X, COSMO-SkyMed, and Radarsat-2, the incidence angle range is mostly within 15–60 [29]. In area 0-1-2-6, when the slope is larger than the incidence angle, layover occurs. The result is that the side-looking viewing property of SAR imagery causes integration of multiple signals with the same range and Doppler frequency within a single resolution cell [30]. In area 6-3-4, when the SAR signal cannot reach the earth surface, shadow occurs. Here, area 6-3 is called the activate shadow area and area 3-4 is called the passive shadow area.

1) Foreshortening: Fig. 5 shows the illustration of foreshortening. In Fig. 5(a), A and B are the points on the earth surface, BE⊥AE, AM⊥AB. Plane A-E parallels to the horizontal plane. Plane S-a-b is the imaging plane. L is the slope length between A and B and imaged in ab. ΔL is the slant range. S is the sensor position. $\sigma 0$ is the front slope angle and $0 \leq \sigma 0 \leq 90°$. α is the local incidence angle. β is the

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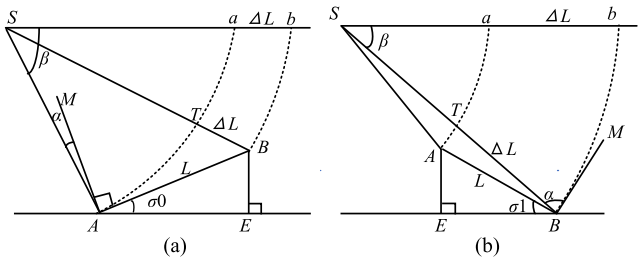


Fig. 5. Illustration of foreshortening. (a) Foreshortening on the front slope. (b) Foreshortening on the back slope.

depression angle. In Fig. 5(b), $\sigma 1$ is the back slope angle and $0 \le \sigma 1 \le 90°$. Plane E-B parallels to the horizontal plane, AE⊥EB, BM⊥AB. The meaning of other parameters is the same as in Fig. 5(a).

1) In Fig. 5(a), foreshortening occurs on the front slope. The horizontal distance between AE is marked as L_{AE}, for the local area, $L_{\mathrm{AE}} = L \cdot \cos(\sigma 0)$. The slant range and slope length satisfy $\Delta L = L \cdot \sin(\alpha)$ [32]. Where $\alpha = 90° - (\beta + \sigma 0)$, $\beta + \sigma 0 \le 90°$. The imaging plane distance between ab is marked as L_{ab}, $L_{\mathrm{ab}} = \Delta L$. The horizontal distance is related to the number of actual pixels in object space. The imaging plane distance is related to the number of image pixels in image space. Therefore, the ratio of the horizontal distance to the imaging plane distance can be regarded as the ratio of the number of actual pixels to an image pixel. L_{AE} is compressed to L_{ab} (ΔL), expressed as follows:

$$f_F(\sigma 0, \beta) = \frac{L_{AE}}{\Delta L} = \frac{L \cdot \cos(\sigma 0)}{L \cdot \sin(\alpha)} = \frac{\cos(\sigma 0)}{\cos(\beta + \sigma 0)}. \quad (9)$$

The SAR satellite incidence angle range is 15°–60°. Since the sum of the incidence angle and the depression angle is approximately 90° [32], the depression angle β range is approximately 30°–75°. To analyze the change law of $f_F(\sigma 0, \beta)$, set the variables $\sigma 0$ and β to different values ($0 \le \sigma 0 \le 90°$, $30° \le \beta \le 75°$, the interval is set to 0.01°), and the number $N1$ that meets the condition of $\beta + \sigma 0 \le 90°$ is calculated. At the same time, the number of $f_F(\sigma 0, \beta)$ in the interval [0,2), [0,3) and [0,4) (denoted as $N2$, $N3$, and $N4$) is counted, respectively. Therefore, the probability that $f_F(\sigma 0, \beta)$ lies in the interval [0,2), [0,3), and [0,4) can be calculated separately by $(N2/N1)$, $(N3/N1)$, and $(N4/N1)$ [denoted as $P(f_F < 2)$, $P(f_F < 3)$, and $P(f_F < 4)$]. As the condition $\beta + \sigma 0$ changes, different probability values are obtained. The results are as shown in Table I. In Table I, as the sum of β and $\sigma 0$ is smaller, the corresponding probability P is larger. When $\beta + \sigma 0 \le 70°$, $P(f_F < 3) = 1$, the ratio of the horizontal distance to the imaging plane distance is less than 3. So the number of actual pixels in object space corresponding to an image pixel is less than 3.

2) For the back slope, as shown in Fig. 5(b), $\sigma 1 < \beta$ is satisfied when foreshortening occurs. The horizontal distance between EB is marked as L_{EB}, similar to (9)

$$f_B(\sigma 1, \beta) = \frac{L_{\mathrm{EB}}}{\Delta L} = \frac{L \cdot \cos(\sigma 1)}{L \cdot sin(\alpha)} \quad (10)$$

TABLE I

PROBABILITY OF f_F WITH THE CHANGE IN β AND $\sigma 0$

Item	$P(f_F < 2)$	$P(f_F < 3)$	$P(f_F < 4)$
$\beta + \sigma 0 \le 90°$	0.3009	0.5349	0.6615
$\beta + \sigma 0 \le 85°$	0.3417	0.6172	0.7633
$\beta + \sigma 0 \le 80°$	0.4102	0.7294	0.9055
$\beta + \sigma 0 \le 75°$	0.5013	0.8863	1
$\beta + \sigma 0 \le 70°$	0.6346	1	1

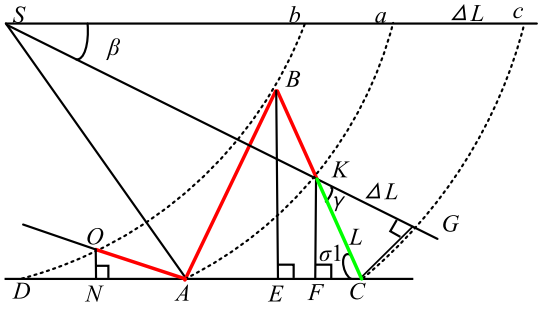


Fig. 6. Illustration of shadow.

where $\alpha = 90° - (\beta - \sigma 1)$. After statistical analysis, the probability P of $f_B < 3$ is 0.9958 ($P(f_B < 3) = 0.9958$). This shows that when foreshortening occurs on the back slope, the number of actual pixels in object space corresponding to an image pixel is basically less than 3.

2) Shadow: In Fig. 4, the shadow is divided into activate shadow area and passive shadow area. For the passive shadow area, area 3-4 in Fig. 4 is similar to area 4-5. Therefore, it is the same as the case of foreshortening.

For the activate shadow area, the details are shown in Fig. 6, part KC. In Fig. 6, KF⊥AC, γ is the angle between the line KC and KG. O is the point on the earth surface, ON⊥AC. L is the slope length between KC and imaged in AC. The meaning of other parameters is the same as Fig. 5(b). For the local area, KG⊥CG can be considered. ΔL represents the slant range between AC.

When active shadow occurs, $\sigma 1 > \beta$ must be met. Therefore, $\gamma = \sigma 1 - \beta$. The horizontal distance between FC is marked as L_{FC}. According to the triangle geometric relationship, $\Delta L = L \cdot \cos(\gamma)$ and $L_{FC} = L \cdot \cos(\sigma 1)$ are established. Therefore

$$\varphi_A(\sigma 1, \beta) = \frac{L_{FC}}{\Delta L} = \frac{L \cdot \cos(\sigma 1)}{L \cdot \cos(\gamma)} = \frac{\cos(\sigma 1)}{\cos(\sigma 1 - \beta)}. \quad (11)$$

The analysis shows that as the variables $\sigma 1$ and β change, $\varphi_A(\sigma 1, \beta)$ takes a value of 0–0.8652, that is, $0 \le \varphi_A(\sigma 1, \beta) \le 0.8652$. Therefore, the number of actual pixels in object space corresponding to an image pixel is less than 3.

3) Layover: Layover can be divided into two situations: the SAR signal can reach the bottom [line SA in Fig. 7(a)] and the other cannot reach the bottom (line SP in Fig. 7(b) and (c). Fig. 7 shows the illustration of layover. In Fig. 7(a), β is the depression angle for point A, β' is the depression angle for point B, ε is the angle between line AS and AO, and $\sigma 0$ is the front slope angle. The meaning of other parameters is the same

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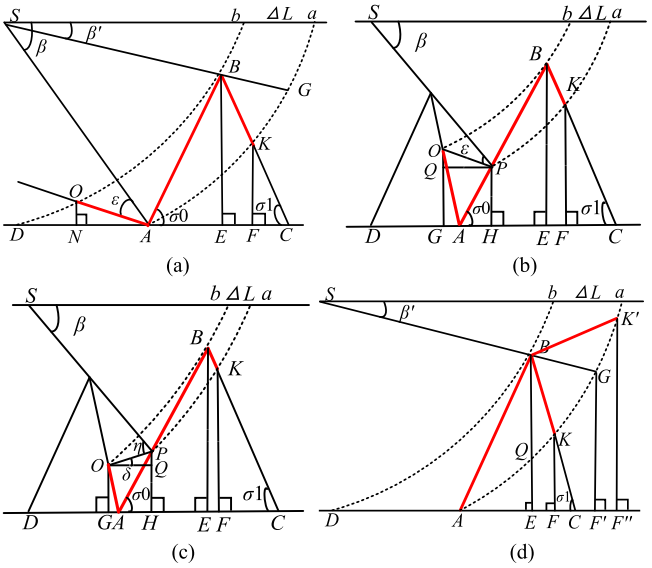


Fig. 7. Illustration of the layover. (a) Signal can reach the bottom. (b) Case 1: Signal cannot reach (bottom). (c) Case 2: Signal cannot reach (bottom). (d) Different situations for BK part.

as in Fig. 6. In Fig. 7(b), β is the depression angle for point P. Here, the SAR signal cannot reach bottom (point A). ε is the angle between line SP and OP, QP⊥OG, OG⊥AC, PH⊥AC. The meaning of other parameters is the same as in Fig. 7(a). In Fig. 7(c), δ is the angle between line OP and OQ, and η is the angle between line OP and SP, OQ⊥PH. The meaning of other parameters is the same as in Fig. 7(b). Fig. 7(d) shows the different situations of BK part.

First, in Fig. 7(a), when the SAR signal can reach the bottom. Obviously, $\varepsilon \leq \beta$. For local area, β' is approximately equal to β, that is, $\beta \approx \beta'$.

In OA part, the following equation relationship can be constructed from the geometric relationship:

$$\begin{cases} L_{\mathrm{NA}} = L_{\mathrm{OA}} \cdot \cos(\beta - \varepsilon) \\ \Delta L = L_{\mathrm{OA}} \cdot \cos(\varepsilon) \end{cases} \tag{12}$$

where L_{NA} means the distance between point N and point A. L_{OA} means the distance between point O and point A. In the subsequent analysis, the distance between two points is expressed in this way and the corresponding symbol is no longer explained.

In AB part, we can construct the following equation:

$$\begin{cases} L_{\mathrm{AE}} = L_{\mathrm{AB}} \cdot \cos(\sigma 0) \\ \Delta L = L_{\mathrm{AB}} \cdot \cos(180° - (\beta + \sigma 0)). \end{cases} \tag{13}$$

In BK part, in fact, this part is the same as part KC in Fig. 6, except that the depression angle is β' at this time. Therefore, a similar relationship can be constructed as follows:

$$\begin{cases} L_{\mathrm{EF}} = L_{\mathrm{BK}} \cdot \cos(\sigma 1) \\ \Delta L = L_{\mathrm{BK}} \cdot \cos(\sigma 1 - \beta'). \end{cases} \tag{14}$$

Taking into account that $\beta \approx \beta'$, (14) can be rewritten as

$$\begin{cases} L_{\mathrm{EF}} = L_{\mathrm{BK}} \cdot \cos(\sigma 1) \\ \Delta L = L_{\mathrm{BK}} \cdot \cos(\sigma 1 - \beta). \end{cases} \tag{15}$$

TABLE II

PROBABILITY OF ψ_A WITH THE CHANGE IN $\sigma 0$ AND $\sigma 1$

Item	$P(\psi_A \geq 2)$	$P(\psi_A \geq 3)$	$P(\psi_A \geq 4)$
$\sigma 0 \leq 90°, \sigma 1 \leq 90°$	0.5730	0.3791	0.2739
$\sigma 0 \leq 80°, \sigma 1 \leq 80°$	0.6716	0.4690	0.3440
$\sigma 0 \leq 70°, \sigma 1 \leq 70°$	0.7819	0.6206	0.4752
$\sigma 0 \leq 60°, \sigma 1 \leq 60°$	0.9070	0.8286	0.7221
$\sigma 0 \leq 50°, \sigma 1 \leq 50°$	0.9559	0.9532	0.9480

Therefore, combining (12), (13), and (15) we can establish the ratio of the horizontal distance to the imaging plane distance as follows:

$$\begin{aligned} &\psi_A(\sigma 0, \sigma 1, \varepsilon, \beta) \\ &\quad = \frac{L_{\mathrm{NA}} + L_{\mathrm{AE}} + L_{\mathrm{EF}}}{\Delta L} \\ &\quad = \frac{\cos(\beta - \varepsilon)}{\cos(\varepsilon)} - \frac{\cos(\sigma 0)}{\cos(\beta + \sigma 0)} + \frac{\cos(\sigma 1)}{\cos(\sigma 1 - \beta)} \end{aligned} \tag{16}$$

where $90° \leq \beta + \sigma 0$, $30° \leq \beta \leq 75°$, $\varepsilon \leq \beta$, $\beta \leq \sigma 1$. Therefore, $15° \leq \sigma 0 \leq 90°$, $30° \leq \sigma 1 \leq 90°$, and $0° \leq \varepsilon \leq 75°$.

Similarly, we set the interval of variables $\sigma 0, \sigma 1, \varepsilon$, and β to 0.01° and calculate the number of $\psi_A(\sigma 0, \sigma 1, \varepsilon, \beta)$ in each interval ([2, $+\infty$), [3, $+\infty$) and [4, $+\infty$)) under the corresponding conditions. In addition, count the total number when the conditions are met, and the probability can be obtained by division. Here, set $\sigma 0$ and $\sigma 1$ to different value ranges and calculate the corresponding $\psi_A(\sigma 0, \sigma 1, \varepsilon, \beta)$. The results are as shown in Table II.

In Table II, as $\sigma 0$ and $\sigma 1$ decrease, the corresponding probability of P gradually increases. When $\sigma 0 \leq 60°$ and $\sigma 1 \leq 60°$, $P(\psi_A \geq 3) = 0.8286$. In this situation, the number of actual pixels in object space corresponding to an image pixel is mostly (probability is 0.8286) no less than 3 when layover occurs. In fact, a slope angle more than 25 is considered as high mountain area in specifications for aerophotogrammetric office operation [33]. In other words, for most terrain conditions, the slope angle will not exceed 25°, so it will not exceed 60°. Therefore, we can consider that the number of actual pixels in object space corresponding to an image pixel is not less than 3 when layover occurs in most cases.

Second, in Fig. 7(b), the SAR signal cannot reach the bottom. In this situation, with S as the center, the concentric circle passing through point B intersects the front mountain at point O, and the elevation of point O is no lower than point P. Therefore, $\varepsilon \leq \beta$.

In fact, the situation in Fig. 7(b) is exactly the same as Fig. 7(a), so it also satisfies equation (16).

Third, in Fig. 7(c), the SAR signal cannot reach the bottom. At this time, there is some difference from the situation in Fig. 7(b). Here, the elevation of point O is no higher than point P. Therefore, $\beta \leq \eta$.

In OA part and AP part: the following equation can be constructed as follows:

$$\begin{cases} L_{\mathrm{OQ}} = L_{\mathrm{OP}} \cdot \cos(\delta) \\ \Delta L = L_{\mathrm{OP}} \cdot \cos(\eta) \end{cases} \tag{17}$$

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TABLE III
PROBABILITY OF ψ_B WITH THE CHANGE IN $\sigma 0$AND $\sigma 1$

Item	$P(\psi_B \geq 2)$	$P(\psi_B \geq 3)$	$P(\psi_B \geq 4)$
$\sigma 0 \leq 90°, \sigma 1 \leq 90°$	0.4430	0.3352	0.2650
$\sigma 0 \leq 80°, \sigma 1 \leq 80°$	0.5197	0.4060	0.3242
$\sigma 0 \leq 70°, \sigma 1 \leq 70°$	0.6369	0.5279	0.4330
$\sigma 0 \leq 60°, \sigma 1 \leq 60°$	0.7977	0.7181	0.6409
$\sigma 0 \leq 50°, \sigma 1 \leq 50°$	0.9305	0.9185	0.8972

where $\delta \leq \sigma 0 \leq 90°$, $\beta \leq \eta \leq 90°$, $\cos(\delta) \geq \cos(\sigma 0)$, and $\cos(\beta) \geq \cos(\eta)$. Therefore, (18) can be drawn

$$\frac{L_{OQ}}{\Delta L} = \frac{\cos(\delta)}{\cos(\eta)} \geq \frac{\cos(\sigma 0)}{\cos(\eta)} \geq \frac{\cos(\sigma 0)}{\cos(\beta)}. \tag{18}$$

Due to

$$L_{GH} = L_{OQ} = L_{GA} + L_{AH}\frac{L_{GA} + L_{AH}}{\Delta L} \geq \frac{\cos(\sigma 0)}{\cos(\beta)}. \tag{19}$$

In PB and BK part: Obviously, these two parts are the same as AB part and BK part in Fig. 7(a). Therefore

$$\frac{L_{HE} + L_{EF}}{\Delta L} = -\frac{\cos(\sigma 0)}{\cos(\beta + \sigma 0)} + \frac{\cos(\sigma 1)}{\cos(\sigma 1 - \beta)}. \tag{20}$$

Therefore, combining (19) and (20), we can establish the ratio of the horizontal distance to the imaging plane distance as follows:

$$\begin{aligned}\psi_B(\sigma 0, \sigma 1, \beta) &= \frac{L_{GA} + L_{AH} + L_{HE} + L_{EF}}{\Delta L} \\ &\geq \frac{\cos(\sigma 0)}{\cos(\beta)} - \frac{\cos(\sigma 0)}{\cos(\beta + \sigma 0)} + \frac{\cos(\sigma 1)}{\cos(\sigma 1 - \beta)}\end{aligned} \tag{21}$$

where $90° \leq \beta + \sigma 0$, $30 \leq \beta \leq 75°$, and $\beta \leq \sigma 1$. Therefore, $15° \leq \sigma 0 \leq 90°$ and $30° \leq \sigma 1 \leq 90°$. To simplify the calculation, we only calculate the minimum of $\psi_B(\sigma 0, \sigma 1, \beta)$.

As the same, probability is calculated for $\psi_B(\sigma 0, \sigma 1, \beta)$ and the result can be seen from Table III. We can draw the same conclusion as in Fig. 7(b) although $P(\psi_A \geq 3) = 0.7181$ when $\sigma 0 \leq 60°$ and $\sigma 1 \leq 60°$.

Fourth, in Fig. 7(a)–(c), calculation of BK part is based on the assumption that $\beta \leq \sigma 1$, more accurately $\beta' \leq \sigma 1$ [Fig. 7(a)]. Fig. 7(d) shows the different situations of BK part.

Based on the condition that β' is a constant and $\beta' \leq \sigma 1$, when $\sigma 1$ changes from β' to 90°, BK part changes from BG to BQ. At the same time, the corresponding horizontal distance of this part changes from EF' to 0. When condition $\beta' \leq \sigma 1$ is not met, the horizontal distance of BK part changes from EF' to EF" because BK part changes from BG to BK'. Obviously, the horizontal distance of BK part is greater than EF'. In this case, the previously calculated probability will increase, but it will not affect the final analysis and conclusion.

4) RFM Method: In the analysis of the geometric characteristics of foreshortening and shadow, the active shadow and the foreshortening occurring on the back slope meet the following condition: the number of actual pixels in object space corresponding to an image pixel is less than 3 when the sum of depression angle and slope angle is less than 70. Passive shadow and foreshortening occurring on the front slope also satisfies the above conditions.

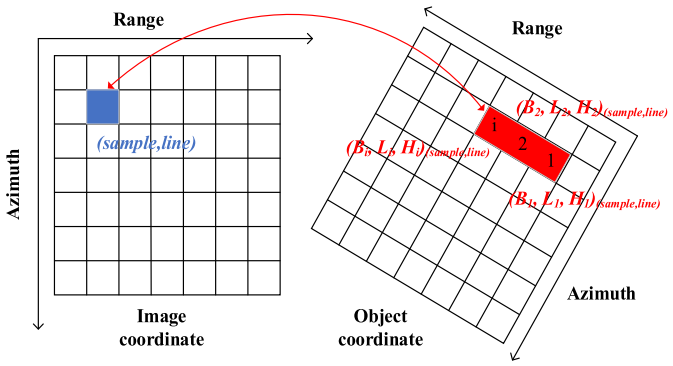


Fig. 8. Relationship between the image coordinate and object coordinate.

For layover, we found that most of the terrain satisfies the number of actual pixels in object space corresponding to an image pixel of no less than 3. Therefore, when the number of actual pixels in object space corresponding to an image pixel is no less than 3, the actual pixels in object space can be approximately determined as layover. Although foreshortening and shadow may be mistaken for layover as the terrain changes, when more than two actual pixels share an image pixel value, the final orthophoto details and texture are difficult to distinguish. So, the area is determined to be layover, which can be further processed. Therefore, the method described in this article determines that layover exists when an image pixel corresponds to more than two (equal to no less than 3) actual pixels in object space. On this basis, combined with RFM, the proposed method is called the RFM method, and the steps are as follows. Fig. 8 shows a simplified diagram of this method.

Step 1: DEM range is determined on the basis of the four corners of the SAR image and the RPCs, and the DEM is resampled at the target resolution.

Step 2: Based on the DEM image coordinate (B, L, H), the corresponding SAR image coordinates (sample, line) are calculated using the RFM, and $(B_i, L_i, H_i)_{(\text{sample,line})}$ corresponding to (sample, line) is recorded ($i = 1, 2, \ldots$).

Step 3: The object coordinates $(B_i, L_i, H_i)_{(\text{sample,line})}$ corresponding to the image coordinates (sample, line) are determined. If $I > 2$, $(B_i, L_i, H_i)_{(\text{sample,line})}$ is regarded as layover and is marked as flag = 2; otherwise, and $(B_i, L_i, H_i)_{(\text{sample,line})}$ is regarded as normal and marked as flag = 1. The mask image is generated by conducting the calculation for each pixel.

Step 4: The mask image is edited by means of morphological "Closing" operation [34] to reduce the "holes" effect, and the LMI is generated.

C. Layover Compensation

Usually, the layover (or shadow) pixel value on the master image is replaced by the normal pixel value on the corresponding position on the slave image, and the following basic

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TABLE IV
BASIC PARAMETERS OF THE TEST AREA IMAGERY

Item	Parameter
Image mode	QPSI
Image resolution (m)	8
Number of images	10
Number of orbits	4

principles apply:

$$\begin{cases} \{[M(i,j) \in A_S] \vee [M(i,j) \in A_L]\} \wedge \\ \left\{\left[S\left(i',j'\right) \in B_S\right] \vee \left[S\left(i',j'\right) \in B_L\right]\right\} \\ \rightarrow R(i,j) = M(i,j) \\ \{[M(i,j) \in A_S] \vee [M(i,j) \in A_L]\} \wedge \left\{\left[S\left(i',j'\right) \in B_N\right]\right\} \\ \rightarrow R(i,j) = S\left(i',j'\right) \\ \{[M(i,j) \in A_N]\} \rightarrow R(i,j) = M(i,j) \end{cases} \tag{22}$$

where $M(i, j)$ is the pixel value in the master image. A_L, A_S, and A_N are the layover, shadow, and normal area in the master image, respectively. $S(i', j')$ shows the corresponding image pixel value of the slave image. B_L, B_S, and B_N are the layover, shadow, and normal area in the slave image, and $R(i, j)$ is the pixel value in the compensated image. The normal area is the remaining part in the image after masking out layover and shadow.

This article mainly focuses on the layover phenomenon, and (22) can be rewritten as

$$\begin{cases} \{[M(i,j) \in A_L]\} \wedge \{[S(i',j') \in B_L]\} \\ \quad \rightarrow R(i,j) = M(i,j) \\ \{[M(i,j) \in A_L]\} \wedge \{[S(i',j') \in B_N]\} \\ \quad \rightarrow R(i,j) = S(i',j') \\ \{[M(i,j) \in A_N]\} \rightarrow R(i,j) = M(i,j). \end{cases} \tag{23}$$

In (23), the normal area is the remaining part in the image after masking out layover. The other parameters have the same meaning as in (22).

IV. EXPERIMENTAL RESULTS AND ANALYSIS

A. Materials and Study Area

In this study, GF-3 QPSI images were used as the test data for the layover compensation experiment. The test area was located between 30.05° and 31.12° latitude and 109.6° and 110.1° longitude. It is a mountainous region of Enshi City in Hubei Province in the central part China. The elevation range is 120–2010 m. Five descending images and five ascending images, a total of ten images, were available for the area, and the acquisition times were from May 31, 2017, to November 2, 2017. Further details of the test area are shown in Table IV and Fig. 9.

B. Geo-Preprocessing

The full-coverage descending images are selected as the MID, and the corresponding ascending images are selected as the SID. The image overlapping rate result is shown in Table V. In this article, layover compensation is performed only between images with overlapping rate greater than 20%.

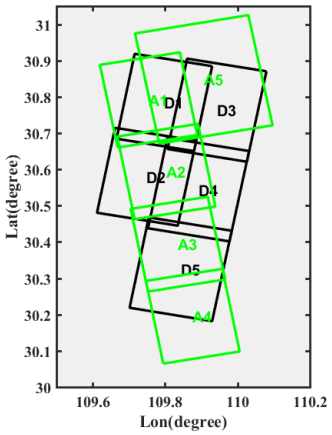


Fig. 9. Image distribution of the test area: D1–D5 represent the descending images that cover the whole area. A1–A5 represent the ascending images that partially cover the test area.

TABLE V
OVERLAPPING RATE OF IMAGES

Master image	Slave image
D1	A1 (77%), A5 (64%)
D2	A2 (70%)
D3	A5 (77%)
D4	A2 (46%), A3 (27%)
D5	A3 (61%), A4 (37%)

TABLE VI
TPS ACCURACY OF BLOCK ADJUSTMENT WITHOUT GCPS

Test Area	TP	Maximum Error (pixel)			RMSE (pixel)		
		x	y	plane	x	y	plane
Enshi	110	-0.694	0.854	0.979	0.297	0.284	0.411

Then, original RPCs are refined by two steps. First, they are refined by geometric calibration. Second, a robust registration method based on structural similarity is adopted to automatically obtain the tie points (TPs) between the images [35]. Then, RPCs are updated again by block adjustment. The registration error in image space of TPs is shown in Table VI. The RMSE of TPs is better than 0.5 pixel and the maximum error is better than 1 pixel, which ensures a seamless geometric mosaic.

C. Automatic Determination of Layover Area

We have described how layover is determined to exist when an image pixel corresponds to more than two actual pixels, and the threshold is set as 2 (TH = 2). To confirm the effectiveness of this method, layover mask was generated by setting different thresholds for different images. In the experiment, one GF-3 QPSI image (image D3 in Fig. 9) and one GF-3 FSII image were selected, which were located in the northern mountainous

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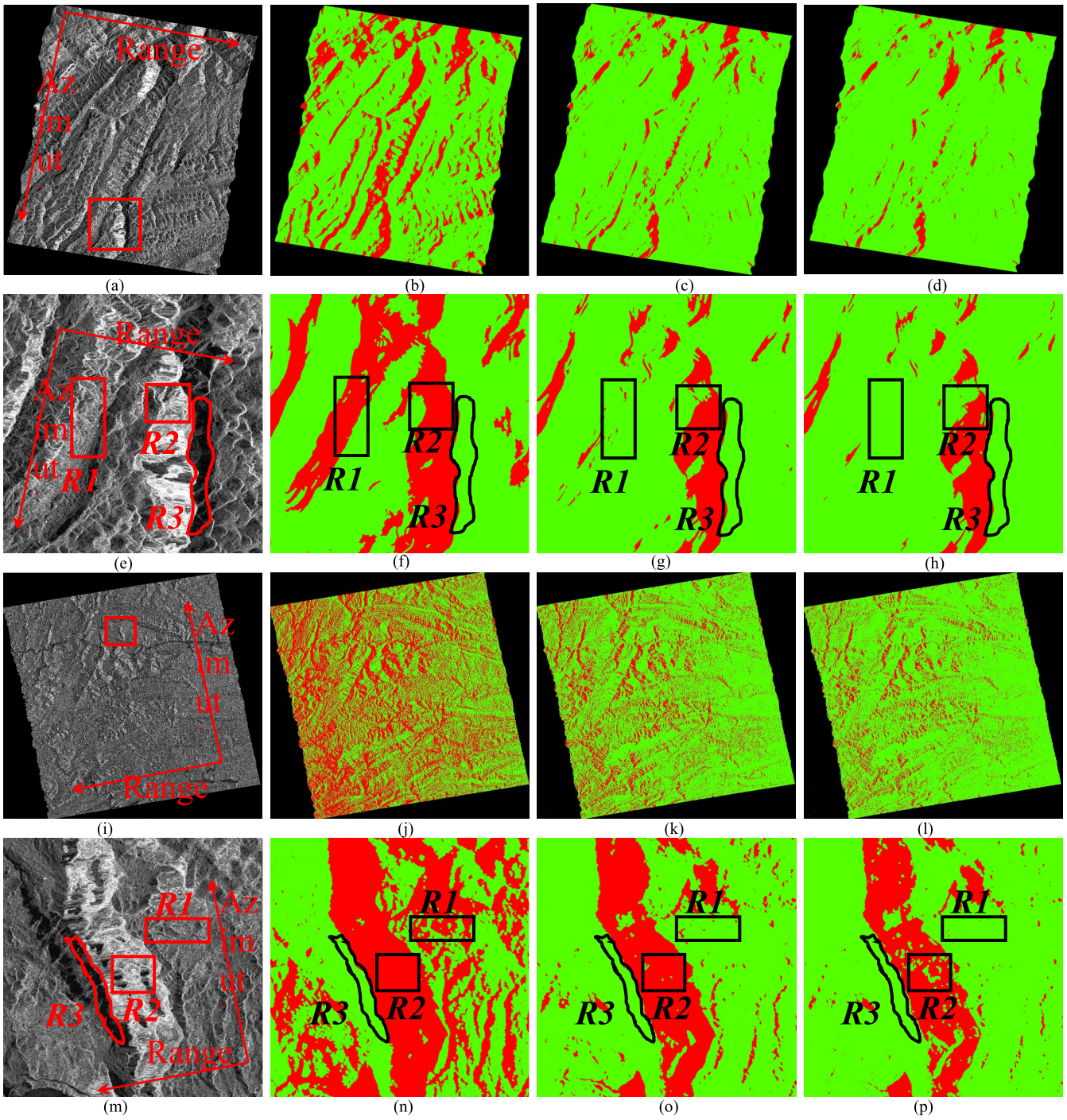


Fig. 10. Different threshold comparison results. (a) GF-3 QPSI orthophoto. (b)–(d) Masks of the area corresponding to (a), and the thresholds are set to TH = 1, TH = 2, and TH = 3, respectively. (e) Enlarged detail of the rectangle in (a), (f)–(h) are the masks of the area corresponding to (e), and the thresholds are set to TH = 1, TH = 2, and TH = 3, respectively. (i) GF-3 FSII orthophoto. (j)–(l) are the masks of the area corresponding to (i), and the thresholds are set to TH = 1, TH = 2 and TH = 3, respectively. (m) Enlarged detail of the rectangle in (i), (n)–(p) are the masks of the area corresponding to (m), and the thresholds are set to TH = 1, TH = 2 and TH = 3, respectively. represents normal area, represents layover area.

area of Hubei Province. Further details of two images are shown in Table VII. The result is shown in Fig. 9.

It can be seen from Fig. 10(b)–(d) that as the threshold increases, the determined layover area gradually decreases. Among them, when the threshold is changed from 1 to 2, the decrease is faster, as shown in Fig. 10(b) and (c). Fig. 10(j)–(l) also satisfies the above conclusion.

In Fig. 10(e), ***R1*** is manually interpreted as a foreshortening area. However, ***R1*** is mainly marked in red, which indicates layover in Fig. 10(f). In Fig. 10(g) and (h), ***R1*** is mainly marked in green, which reflects the actual situation. In Fig. 10(e), ***R2*** is manually interpreted as a layover area. Compared with Fig. 10(g), "holes" appear in this area Fig. 10(h). Although the mask is edited by means

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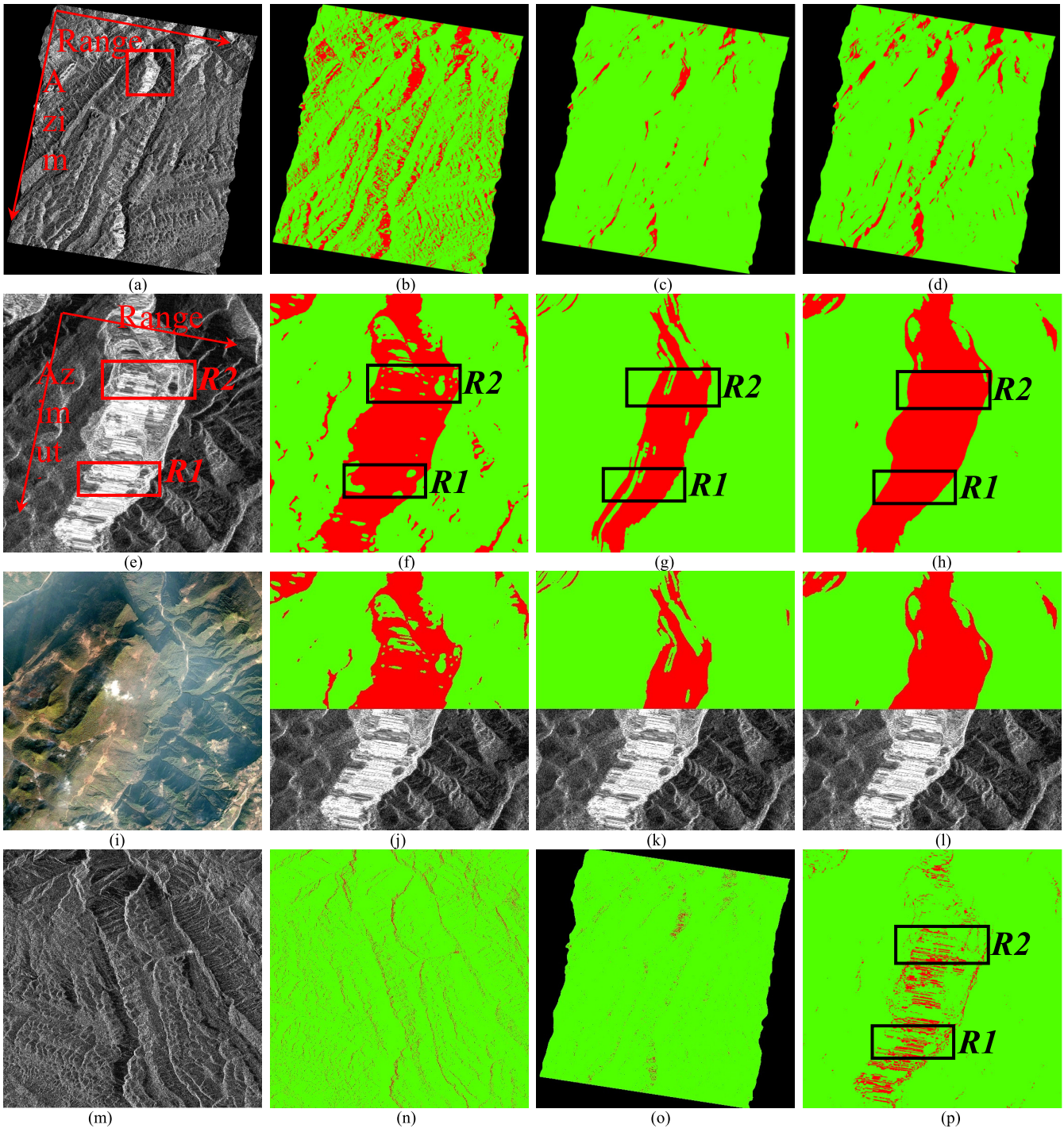


Fig. 11. Results of method comparisons. (a) GF-3 QPSI orthophoto. (b) Gamma software improvement method mask. (c) Literature [7] method mask. (d) RFM method mask. (e) Enlarged detail of the rectangle in (a). (f) Gamma software improvement method mask corresponding to (e). (g) Literature [7] method mask corresponding to (e). (h) RFM method mask corresponding to (e). (i) DOM generated by ZY-3 satellite corresponding to (e). (j)–(l) are (f)–(h) blended into (e), respectively. (m) SAR Image. (n) Gamma software method mask. (o) Gamma software method mask after rectified. (p) Gamma software method mask after rectified corresponding to (e). represents normal area, represents layover area.

of morphological "Closing" operations, it is still not filled completely. R3 is manually interpreted as a shadow area, and it is easy to see that the shadow area is rarely misjudged as the layover area by the RFM method. In Fig. 10(n)–(p), the situation is the same as Fig. 10(f)–(h), so the same conclusion can be drawn.

As can be seen from comparison with Fig. 10, if the threshold value is higher or lower, the obtained mask results will be significantly different from the actual situation. Therefore, the threshold is set as 2 in this article. The experiment uses images with different incidence angles and different viewing directions. It is verified that when the threshold

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TABLE VII
BASIC PARAMETERS OF TWO IMAGES

Item	QPSI	FSII
Incidence angle range	39.453°~40.588°	31.333°~38.154°
Image center	30.764N, 109.917E	30.149N, 110.675E
Image size	5749×5993	28171×26674
Viewing direction	Descending	Ascending
Time	2017.07	2017.10

is 2, the RFM method can effectively determine the layover area.

In Gamma software, several times of the image average value are selected as threshold. In this article, we select three times average as threshold, called the Gamma software method. Fig. 11(n) shows the result of the Gamma software method. It needs to be rectified before performing compensation. In Fig. 11(o), the rectified mask is severely discretized and is difficult to determine the layover area effectively. Therefore, on the basis of the Gamma software method, combined with the morphological "Closing" operation, we called the Gamma software improvement method. Fig. 11(b) shows the result of this method. To better evaluate the performance of the RFM method proposed in Section III-B, Gamma the software improvement method and a method described in the literature [7] were selected for comparison.

In Fig. 11(f), although "Closing" operation is conducted, there are still many "holes" in the middle of R1 and R2. In Fig. 11(g), R1 and R2 still have holes, but they are not found in Fig. 11(h), which is the closest to the actual situation. As shown in Fig. 11(j)–(l), the three methods yield good positioning accuracy results, but as shown in Fig. 11(g), because of the error in the slope and other variables, the layover range and the actual range are relatively small. The comparison results suggest that the RFM method is effective.

D. Layover Compensation on Two Images

To further confirm the effectiveness of the RFM method, layover compensation experiments were designed using three different methods.

In those experiments, D3 and A5 in Fig. 9 were selected as the master image and the slave image respectively. Here, image A5 and D3 absolute and relative geometric accuracy was high by geoprocessing. Then, orthorectification was carried out, and layover masks were generated by the RFM method, Gamma software improvement method, and literature [7] method at the same time. Finally, layover compensation was conducted. Table VIII shows the statistical results of the number of layover pixels before and after compensation in the overlapping area by different methods in the master image; meanwhile, the compensation rate, mean, and standard deviation (STD) after compensation were calculated. The mean and standard deviation of the overlapping area in the master image before compensation were 131.3 and 128.2, respectively. The compensation rate is defined as [10]

$$\mathrm{Rate}_{\mathrm{comp}} = \frac{\mathrm{CNT}_{\mathrm{comp}}}{\mathrm{CNT}_{\mathrm{before}}} \times 100\% \tag{24}$$

TABLE VIII
RESULTS OF DIFFERENT METHODS AFTER COMPENSATION

Item	Gamma software improvement method	Literature [7] method	RFM method
Layover pixels	4117419	712099	2038897
Compensated pixels	4102566	702366	2018693
Compensation rate	98.6%	99.6%	99.0%
Mean after compensation	99.3	124.3	115.1
STD after compensation	69.7	117.8	104.7

where $\mathrm{Rate}_{\mathrm{comp}}$ is the compensation rate,$\mathrm{CNT}_{\mathrm{comp}}$ is the number of compensated pixels in the overlapping area, and $\mathrm{CNT}_{\mathrm{before}}$ is the number of layover pixels in the overlapping area. In Table VIII, the compensation rates for the three different methods were better than 98%, which indirectly indicates the feasibility of the use of the ascending and descending images for layover compensation. The Gamma software improvement method detected more layover pixels, and the mean and standard deviation were smaller after compensation. However, because of the larger influence of pixel values, there were more misjudgments, and "holes" were difficult to eliminate, which affected further application. Literature [7] method was affected by slope and other variables, resulting in missed layover judgment. The layover pixels obtained by the RFM method is approximately half of the Gamma software improvement method and three times of literature [7] method. In the RFM method, after compensation, the mean and standard deviation decreased by approximately 12.3% and 18.3%, respectively. The layover was effectively compensated at the same time. Fig. 12 shows the compensation results by three different methods.

In Fig. 12, due to many "holes" in the Gamma software improvement method mask, there are lots of "spots" in the hole positions on the compensated image, as shown in Fig. 12(f). Literature [7] method determinates few layover pixels, and therefore, the compensation effect is not obvious in Fig. 12(g). It is found that using the RFM method to generate mask, the final layover compensation effect is obvious.

E. Regional Layover Compensation

To verify the performance of the proposed layover compensation method for regional imagery, two different experiments were designed. First, regional DOM was generated by the proposed method in this article. Second, regional DOM was generated without layover compensation.

In geo-preprocessing, the RMSE of TPs is better than 0.5 pixel, which satisfies the requirement of a geometric seamless mosaic. Then, using refined RPCs, orthorectification was carried out for all images, and LMI was generated. Finally, layover compensation was carried out and regional compensated DOM was obtained through mosaic, as shown in Fig. 13(a). Without layover compensation, all the master images were orthorectified, and then we obtained regional SAR DOM by mosaic. The result is shown in Fig. 13(b).

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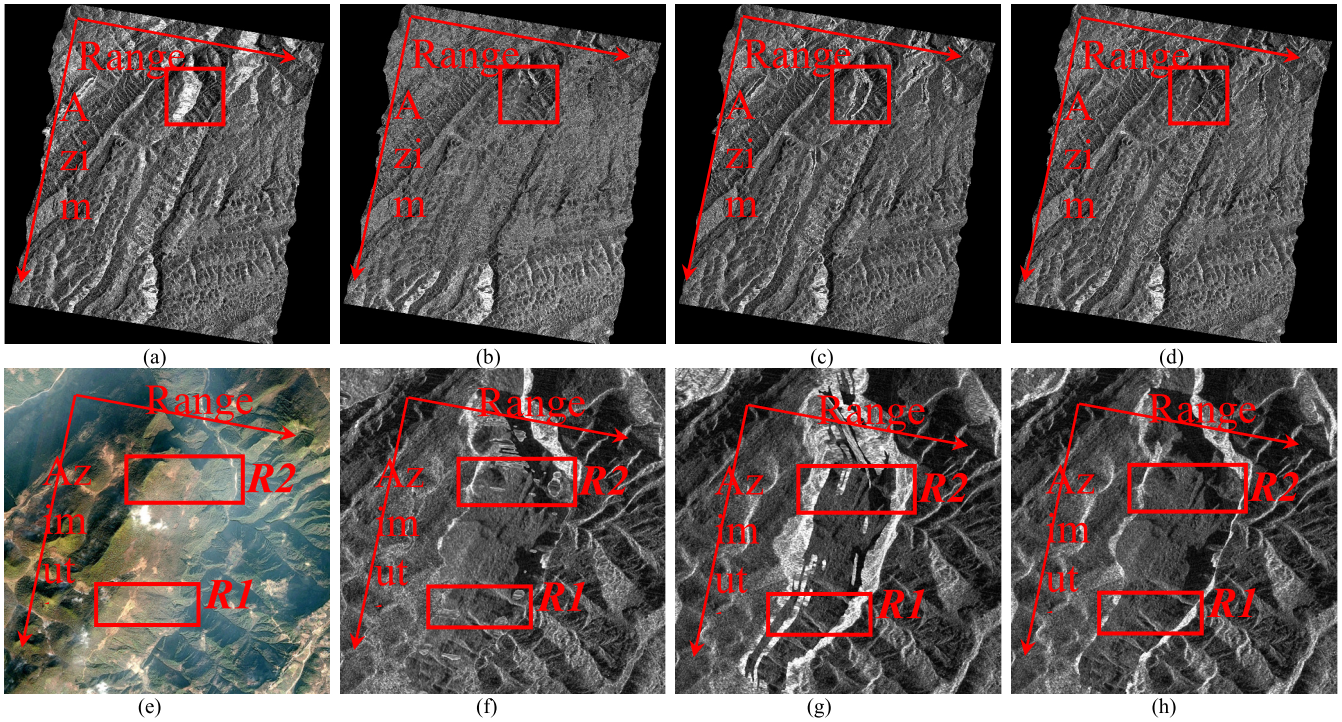


Fig. 12. Different methods compensation result. (a) GF-3 QPSI orthophoto. (b) Gamma software improvement method. (c) Literature [7] method. (d) RFM method. (e) Enlarged detail of the rectangle in (a). (f) Enlarged detail of the rectangle in (b). (g) Enlarged detail of the rectangle in (c). (h) Enlarged detail of the rectangle in (d).

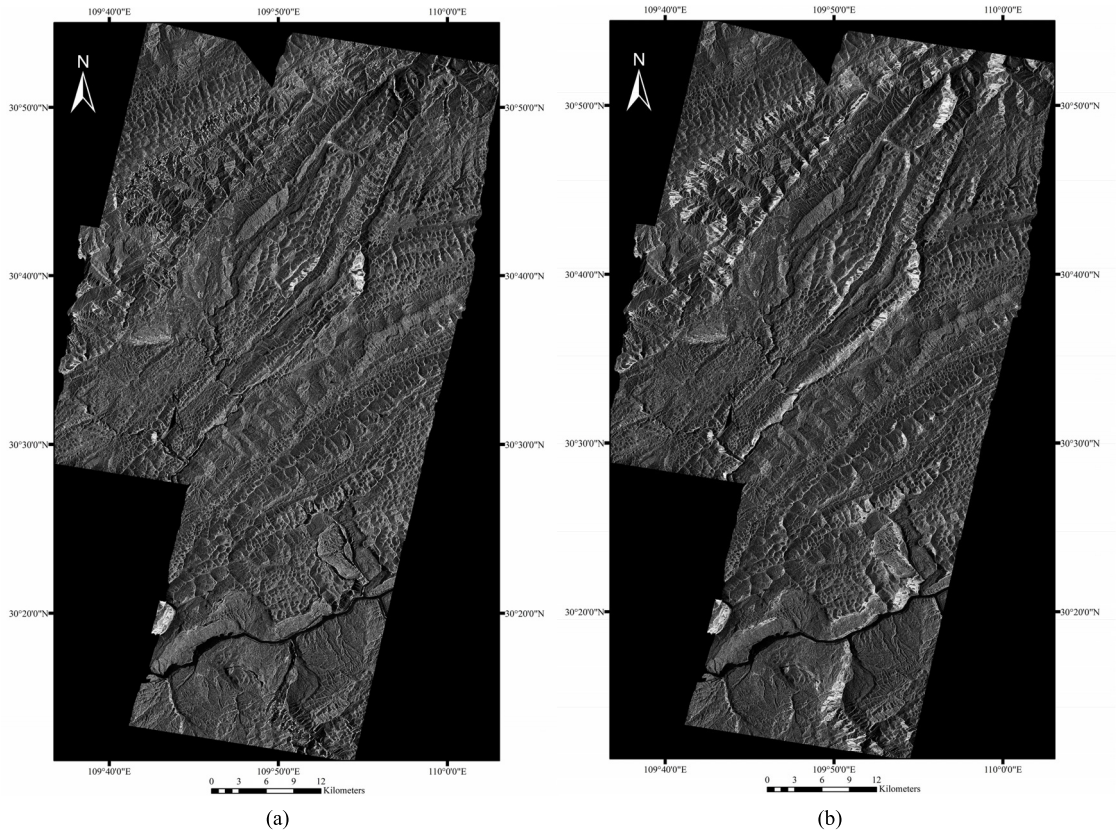


Fig. 13. Comparison of regional DOM before and after compensation. (a) After compensation. (b) Before compensation.

For verifying the accuracy of the compensated SAR DOM, 12 independent check points (ICPs) were obtained from DOM and DEM. DOM and DEM were generated by ZY-3 satellite stereoscopic images with a spatial accuracy of 5 m in plane and 2 m in elevation. Considering that the features of the SAR image are not easy to identify, all ICPs were selected

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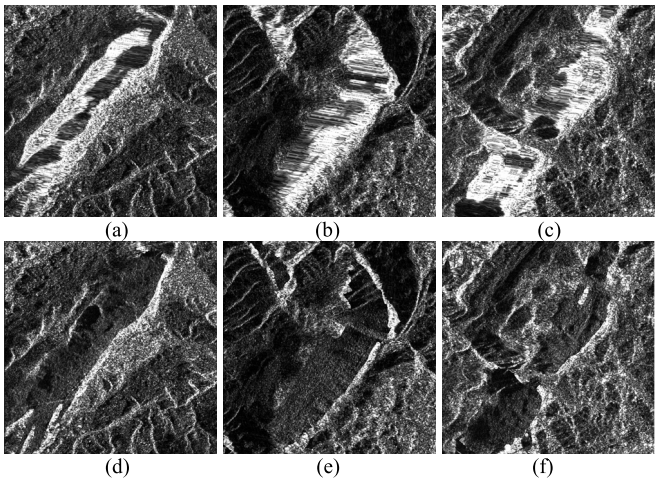


Fig. 14. Proposed method. Comparison of local DOM before and after compensation. (a)–(c) Local images before compensation. (d)–(f) Respective images after compensation.

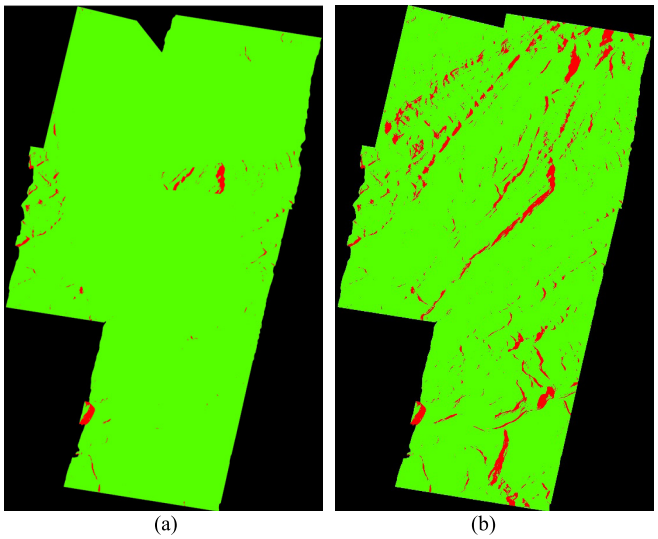


Fig. 15. Comparison of regional mask before and after compensation. (a) After compensation. (b) Before compensation.

from prominent ground features, such as road intersections or water bodies. Therefore, the manual measurement accuracy of the ICPs was about ±1 pixel. The RMSE of 12 ICPs was 8.67 m in the east direction, 7.80 m in the north direction, and 11.45 m in plan. Relative to 8-m resolution of the QPSI model, the accuracy is better than 1.5 pixels in image space.

The layover shown in Fig. 13(b), which appears as white portions in the image, was significantly reduced after compensation, as shown in Fig. 13(a). Fig. 14 shows a comparison of the details before and after compensation. After the highlighted white portions are replaced by the effective pixel value, the texture structure of the ground object is clear. The effect of compensation is obvious. In this article, the number of layover pixels in Fig. 13(a) and (b) was counted, respectively. The number of the remaining layover pixels after compensation was 1 080 303 [in Fig. 13(a)], and the number of pixels before compensation was 6 959 656 [in Fig. 13(b)]. Therefore, a total of 5 879 353 pixels were compensated. In other words, 84.5% of the layover pixels are effectively compensated. Fig. 15 shows the regional layover mask before and after compensation.

V. Conclusion

In this study, GF-3 QPSI images were used to test the proposed layover compensation method for spaceborne SAR imagery. The following conclusions can be drawn.

It was feasible to generate layover mask by a new determination method based on RFM, the mask result was closer to the actual situation, and compensation result showed that the compensation rate reached to 99.0%. Second, geometric calibration ensures that GCPs are not necessary on the process of geoprocessing and block adjustment ensures relative geometric accuracy, which equals the accuracy of a seamless geometric mosaic and provides technical assurance for production of regional DOM. By conducting regional layover compensation experiment, a regional DOM was obtained, and the texture information of the layover region increased obviously. Through verification, the geometric accuracy of regional DOM reached to 11.45 m. Besides, 84.5% of the layover pixels on regional DOM were effectively compensated. The proposed regional layover compensation method can not only ensure high accuracy SAR DOM but also fix missing pixels of SAR DOM. This method can promote the further application of SAR image for large area.

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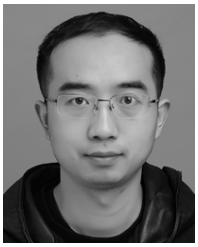
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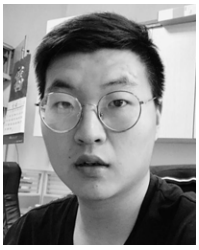
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